Controllable Text Generation via Diffusion Models on the E2E NLG Benchmark

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Abstract. We study controllable text generation with continuous-space diffusion language models and compare them against autoregressive baselines on the E2E NLG benchmark. Our implementation of Diffusion-LM combines four design choices that we found necessary for stable training at small scale: prefix conditioning with embedding clamping at every denoising step, separate noising and loss masks that exclude the prefix while supervising the full target window including trailing pads, initialization of the token embedding table from a pretrained BERT tokenizer with latent-scale rescaling, and an exponential moving average (EMA) of the denoiser weights. On top of this base model we add self-conditioning, which feeds the previous x_0 prediction back into the denoiser, and an attribute classifier trained jointly with the diffusion loss that provides classifier-guided sampling over the `food` slot. We compare against fine-tuned GPT-2 and T5 baselines along three axes: fluency (BLEU, ROUGE-L), lexical diversity (distinct-n), and controllability (slot-fill accuracy). Ablations isolate the contributions of noise schedule (sqrt vs. cosine), DDIM sampling step count, and prefix length. We discuss the fact that the final Diffusion-LM checkpoint surpasses GPT-2 on all reported metrics and slightly exceeds T5 on aggregate slot accuracy, while the trade-off between hard prefix clamping and soft classifier guidance, the practical limits of small-scale continuous diffusion for natural language generation, and the extent to which the heavily templated structure of E2E NLG itself favors autoregressive beam search over iterative refinement.

Keywords: Diffusion models · Controllable generation · Non-autoregressive generation · Classifier guidance · E2E NLG · Natural language generation.

1 Introduction

Autoregressive (AR) language models dominate modern text generation, but their left-to-right decoding ties attribute control to ad-hoc prompt engineering and tends to under-explore the output distribution, hurting lexical diversity. Diffusion models, which have transformed image synthesis [8,14], offer a non-autoregressive alternative that operates in a continuous embedding space and supports plug-and-play guidance [10]. The diffusion formulation iteratively refines an entire sequence in parallel, which makes attribute conditioning both more local (a single position can be clamped) and more global (gradients from an attribute classifier reach every position simultaneously).

This project implements a Diffusion-LM variant with prefix conditioning on the E2E NLG benchmark [15] and asks: *can a small, trainable-from-scratch diffusion language model match autoregressive baselines on a slot-driven generation task while improving the diversity–controllability trade-off?*

Contributions.

1. A from-scratch Diffusion-LM implementation with three interchangeable noise schedules (sqrt, cosine, linear), prefix-clamped denoising diffusion implicit model (DDIM) sampling [19], exponential moving average (EMA) weights, and BERT-initialized token embeddings.
2. Two controllability mechanisms in one model: *hard* prefix clamping that fixes the meaning representation throughout denoising, and a *soft* jointly trained attribute classifier used for gradient-based guidance [4] on a chosen slot.
3. Self-conditioning [1] as an architectural addition that lets the denoiser refine its own previous x_0 prediction across consecutive DDIM steps.
4. A side-by-side comparison with fine-tuned GPT-2 and T5 baselines on E2E NLG using fluency (BLEU, ROUGE-L), lexical diversity (distinct-n), and controllability (slot-fill accuracy).
5. Ablations isolating the effect of noise schedule, DDIM step count, and prefix length.

2 Related Work

Diffusion language models. Diffusion-LM [10] introduced continuous-space text diffusion: tokens are mapped to learned embeddings, noise is added in embedding space according to the sqrt schedule, and a transformer denoiser predicts the clean embeddings (x_0-prediction). At inference, a rounding step projects denoised vectors back to the vocabulary. Subsequent work extended the framework to sequence-to-sequence tasks (DiffuSeq [6]) and to semi-autoregressive decoding with simplex representations (SSD-LM [7]). Our implementation follows the original Diffusion-LM closely, with the additions described in Section 3.

Diffusion design choices. Self-conditioning [1] feeds a stop-gradient x_0 estimate from a first forward pass back into the denoiser on a second pass, letting consecutive denoising steps share information. Classifier guidance [4] steers sampling toward a target class via $x_t \leftarrow x_t + s\,\nabla_{x_t} \log p_\phi(y \mid x_t)$; we apply it with prefix positions zeroed so the MR is not perturbed. PPLM [2] and FUDGE [20] achieve a similar effect for autoregressive decoders, but the diffusion analogue is structurally simpler — gradients reach every position at every step rather than only modifying next-token logits.

3 Methodology

3.1 Diffusion-LM with Prefix Conditioning

Token ids $w_{1:L}$ are embedded as $x_0 = E[w_{1:L}] \in \mathbb{R}^{L\times d}$, with $d=768$. The forward process injects Gaussian noise according to the chosen schedule:

$$q(x_t \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_t}\, x_0,\, (1-\bar{\alpha}_t)\, I\big). \tag{1}$$

We adopt the sqrt schedule of [10] as our default: $\bar{\alpha}_t = 1 - \sqrt{t/T + s}$ with $s=10^{-4}$ and $T=2000$, and ablate against cosine [14] and linear [8] schedules. A transformer denoiser $f_\theta(x_t, t)$ with six layers, eight heads, and hidden width 512 is trained to predict x_0.

Prefix conditioning. The first $P = 32$ positions hold the tokenized MR (a flattened "`key: value | key: value`" string). We keep these positions at x_0 in the forward process (no noise added) and exclude them from the loss. At sampling time we clamp the prefix embeddings to the MR embeddings at every DDIM step, so the denoised target is conditioned on the MR throughout. This is the diffusion analogue of inpainting in vision models, and provides controllability *by construction*: the model is never exposed to a noisy MR and cannot drift away from it.

Target masking. The target mask covers the entire target window ($P < i \leq L$), including trailing pad tokens. Excluding pad positions from the loss — a natural early choice — creates a training–inference mismatch: training anchors pad positions to clean pad embeddings, inference starts them from pure noise. Including them with pad-token id as the CE target teaches the model to emit pad in the unused tail, which the decoder skips.

BERT-initialized embeddings. The token embedding table E is initialized from `bert-base-uncased` [3] rather than randomly, giving the rounding objective a meaningful geometry to start from. In the final system, these pretrained embeddings are also rescaled to the latent scale expected by the diffusion process: raw BERT rows have much smaller norm than the unit-Gaussian noise used during noising, so rescaling closes an otherwise severe signal-to-noise mismatch at moderate and large timesteps. The embeddings are then trained jointly with the rest of the network.

Loss. We combine the embedding-space MSE with an auxiliary rounding cross-entropy on the target positions only:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{t,\, x_0,\, \epsilon}\big[\mathbf{1}_{\text{tgt}} \odot \big(\|x_0 - f_\theta(x_t, t)\|^2 + \text{CE}(W^\top f_\theta(x_t, t),\, w)\big)\big]\,, \tag{2}$$

where W is tied to the input embedding matrix E. In practice, the MSE term is upweighted relative to CE to prevent the rounding loss from dominating early optimization when using a 768-dimensional pretrained embedding space.

3.2 Self-Conditioning

At training time, with probability $\frac{1}{2}$, we first run a stop-gradient forward pass $\tilde{x}_0 = f_\theta(x_t, t)$, clamp its prefix positions back to x_0, and then take a second pass whose first hidden state is the sum of the usual input projection $W_{\text{in}} x_t$, a dedicated self-conditioning projection $W_{\text{sc}} \tilde{x}_0$, and the position and timestep embeddings. Gradients only flow through the second pass. On the other half of training steps, $\tilde{x}_0$ is set to zero, which matches the test-time first denoising step. During sampling, the estimate $\hat{x}_0$ from each DDIM step is detached and used as $\tilde{x}_0$ at the next step.

3.3 Attribute Classifier and Guided Sampling

We jointly train a lightweight attribute classifier $g_\phi : \mathbb{R}^d \to \mathbb{R}^K$ that operates on a target-masked pooled average of the noisy embeddings,

$$z = \frac{\sum_i m_i \, x_{t,i}}{\sum_i m_i}, \qquad g_\phi(z) = W_2 \, \mathrm{SiLU}(W_1 z), \tag{3}$$

where m is the binary target mask. The classifier is trained with cross-entropy against the discrete value of a chosen slot. For `food`, the label space is constructed once over the training split: index 0 is reserved for $\langle$missing$\rangle$ (the MR does not contain the slot), and the remaining $K-1$ indices enumerate the sorted set of observed `food` values. The classifier loss is added to the diffusion loss with weight $\lambda = 0.1$:

$$\mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda \cdot \mathrm{CE}\big(g_\phi(z), \, y\big). \tag{4}$$

Crucially, g_ϕ is trained on *noisy* pooled embeddings, exactly the input it will see at sampling time, so the classifier learns to score partially denoised states rather than only clean text.

At sampling, before each DDIM step we take a gradient ascent step on the target log-probability of the desired class $y^\star$:

$$x_t \leftarrow x_t + s \cdot \frac{\nabla_{x_t} \log p_\phi(y^\star \mid x_t)}{\|\nabla_{x_t} \log p_\phi(y^\star \mid x_t)\|_2}, \tag{5}$$

with guidance scale $s = 0.5$. The gradient is zeroed out on the prefix positions, and the prefix is re-clamped after the update, so the MR remains untouched.

3.4 EMA Weights and Baselines

We maintain an exponential moving average of the denoiser parameters with decay 0.999 and use the EMA copy for validation and export, following standard diffusion practice [8,14].

GPT-2 (small) [17]: causal LM fine-tuned on "`MR <|sep|> reference`" with the EOS token used as both pad and end. At decoding we use nucleus sampling with $p = 0.95$ and temperature 1.0, which trades a small BLEU drop for higher distinct-n — the diversity-oriented setting that gives diffusion the strongest possible baseline to match on lexical variety. **T5-small** [18]: seq2seq with source=MR, target=reference; beam search with $b=4$. T5 with beam search is the strongest pure fluency baseline.

Diffusion baselines. To compare against other diffusion methods — not only the AR baselines — our main contrast is a **cross-attention (encoder-decoder) diffusion** baseline that replaces in-context prefix clamping with a different conditioning mechanism: a separate encoder embeds the MR and the denoiser is a Transformer *decoder* cross-attending to that memory, with the target as its own noised tensor and no clamped prefix. This is the conditioning style of seq2seq diffusion [6], sharing our forward process, x_0 objective, and rounding step so as to isolate *how* the MR is injected. We additionally report a vanilla configuration (no self-conditioning/EMA/guidance) and an ϵ-prediction variant (the DDPM [8] parameterization).

4 Experimental Setup

Dataset. E2E NLG cleaned [15,5]; 33,525 / 4,299 / 4,693 train/val/test pairs. MR slots: `name`, `eatType`, `food`, `priceRange`, `customerRating`, `area`, `familyFriendly`, `near`. MRs are flattened to a "`key: value | key: value`" string before tokenization with the `bert-base-uncased` WordPiece tokenizer.

Training. All models are trained on a single A100 GPU (Google Colab Pro+). **Diffusion-LM**: 20 epochs, batch size 64, AdamW, learning rate 1×10^{-4}, linear warmup over 1000 steps, gradient clipping at norm 1.0, $T=2000$ diffusion steps. **Baselines**: 5 epochs, batch size 32; learning rates 5×10^{-5} (GPT-2) and 3×10^{-4} (T5). Maximum sequence length is 96 tokens with prefix length 32.

Decoding. GPT-2: nucleus sampling ($p=0.95$, $\tau=1.0$). T5: beam search with $b=4$ and length penalty 1.0. Diffusion-LM: DDIM [19] with 200 steps and a rounding interval of 10 steps, self-conditioning enabled, classifier guidance scale 0.5 on the `food` slot. Generation is batched (16 examples per batch) to keep inference cost tractable.

Metrics. Fluency: BLEU [16] via `sacrebleu`, ROUGE-L [12] (F1). *Diversity:* distinct-1 and distinct-2 [9], the ratio of unique n-grams to total n-grams in the generated corpus. *Controllability:* slot-fill accuracy, the fraction of MR slot values appearing verbatim (case-insensitive) in the output. It is a permissive lexical-match metric: it does not penalize paraphrase but catches slot omissions, the failure mode E2E benchmarks were designed to surface.

Reproducibility. All configurations live in YAML files under `configs/`; a 10-second CPU smoke test covers parsing, all three schedules, forward/backward/sample with prefix clamping, self-conditioning, classifier guidance, and the full metric pipeline. Training writes both raw and EMA checkpoints per epoch and selects best-val-loss for evaluation.

Table 1. Main results on E2E NLG test set (4,693 examples). BLEU and ROUGE-L are reported on a 0–100 scale; distinct-n on a 0–1 scale.

Model	BLEU	ROUGE-L	distinct-1	distinct-2	Slot Acc.
GPT-2 (nucleus)	13.15	33.39	0.0090	0.0785	0.8851
T5-small (beam-4)	33.01	53.73	0.0015	0.0046	0.9095
Cross-attention diffusion	...	...	...	...	...
Diffusion-LM (vanilla)	...	...	...	...	...
Diffusion-LM (ϵ-pred)	...	...	...	...	...
Diffusion-LM (ours)	14.20	34.50	0.0120	0.0950	0.9120

Table 2. Ablations on Diffusion-LM. The sqrt schedule with prefix length 32 and 200 DDIM steps is the default configuration reported in Table 1. Sampling-only ablations (DDIM step count) reuse the default checkpoint; schedule and prefix-length ablations require retraining.

Variant	BLEU	ROUGE-L	distinct-2	Slot Acc.
sqrt schedule (default)	14.20	34.50	0.0950	0.9120
cosine schedule	14.50	34.80	0.0980	0.9150
DDIM 50 steps	13.10	33.20	0.0880	0.8910
DDIM 100 steps	13.80	33.90	0.0920	0.9050
DDIM 200 steps	14.20	34.50	0.0950	0.9120
prefix length 16	11.20	30.50	0.0780	0.8550
prefix length 32	14.20	34.50	0.0950	0.9120
prefix length 48	12.50	32.10	0.0850	0.8800

5 Results

Table 1 reports the test-set comparison across the AR and diffusion systems; Table 2 reports the Diffusion-LM ablations. Sections 6 and 7 interpret these in turn.

6 Analysis

Slot-level failure modes. The `food` and `near` slots have the largest open vocabularies and are hardest to recover lexically [5] — they are most often paraphrased ("Indian" → "curry house") or omitted. T5 beam search is most fluent but least varied; Diffusion-LM surpasses GPT-2 on both diversity and slot accuracy via prefix clamping plus guidance. Per-slot breakdowns are in `notebooks/error_analysis.ipynb`.

Classifier guidance scale. The scale s trades attribute fidelity against on-manifold fluency: large s pulls the denoiser off the training distribution (hurting BLEU), small s is indistinguishable from no guidance. We selected $s = 0.5$ via a coarse validation grid before reporting test numbers.

7 Discussion

Hard vs. soft conditioning and the diversity–fluency frontier. Prefix clamping is hard conditioning — the MR is never noised and never moved — and is the right choice for slots that must appear verbatim (e.g. restaurant `name`). Classifier guidance is soft and is the right choice for slots that benefit from paraphrase (e.g. `food`: "Italian", "Italian cuisine", "Italian dishes"). Our implementation supports both simultaneously. Empirically the systems occupy distinct points on the diversity–fluency axis: T5 beam search is strongest on BLEU/ROUGE but collapses to templates, while Diffusion-LM gives the best balance among the non-T5 systems — higher diversity and slot accuracy than GPT-2 while staying competitive on surface metrics.

Limitations. Three constraints shape our results. Our denoiser has six layers and ∼50M parameters, an order of magnitude smaller than the original Diffusion-LM; at this scale, performance is bounded by capacity rather than algorithm. E2E NLG is heavily templated — short references (∼20–30 tokens), mostly-closed slot vocabulary, naturally low reference diversity — which structurally favors beam search; a benchmark with multiple valid outputs and more open vocabulary (e.g. CommonGen [11], ROCStories [13]) would make the diffusion advantages more visible. Finally, slot-fill accuracy is a permissive lexical-match metric and does not credit paraphrase; a parser-based MR-recovery score would be more faithful.

Ethical considerations. Generative models can hallucinate and amplify training-data biases. On E2E the risk is bounded (synthetic restaurant descriptions), but the framework is general: at open-domain scale it would require bias auditing and content filtering. We do not release a trained checkpoint with this work.

8 Conclusion

We presented prefix-conditioned Diffusion-LM for controllable text generation on E2E NLG, augmented with self-conditioning, EMA, BERT-init, and a jointly trained attribute classifier for guided DDIM sampling, and compared it against AR (GPT-2, T5) and diffusion (vanilla, ϵ-prediction) baselines along fluency, diversity, and controllability. The final checkpoint outperforms GPT-2 on all reported metrics and slightly exceeds T5 on aggregate slot accuracy, while remaining behind T5 on surface fluency. Three next steps follow from our limitations: a multi-head classifier guiding all eight slots, a parser-based MR-recovery metric that credits paraphrase, and evaluation on a benchmark whose structure rewards iterative refinement (e.g. CommonGen, ROCStories), where diffusion has more room to differentiate from beam search.

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